

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I (Naveen ms / 192371002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario** ○ Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements** ○ List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

3. **Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
 - Develop an SRS document containing:
 - Introduction
 - Problem Description
 - Scope of the System
 - Functional Requirements
 - Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
5. **Identify Assumptions and Constraints**
 - List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
 - Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.

Software Requirement Specification	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or
(SRS) Documentation	organized, and includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

Software Requirement Analysis (SRA) and Software Requirement Specification (SRS)

Project Title: EV Charging Station Management System

1. Study and Analyze the Scenario

Problem Statement

The increasing adoption of Electric Vehicles (EVs) has created a demand for an efficient charging infrastructure. Many EV users face difficulties in locating nearby charging stations, checking charger availability, booking charging slots, making secure payments, and monitoring charging

progress. The current process is often manual or fragmented across different platforms, leading to delays and inconvenience.

The proposed EV Charging Station Management System is a web-based application that enables users to locate charging stations, reserve charging slots, make online payments, and monitor charging sessions. It also provides administrators with tools to manage charging stations, users, bookings, and payments efficiently.

Business Domain

Electric Vehicle (EV) Infrastructure and Smart Transportation

Stakeholders

Primary Stakeholders

- **EV Vehicle Owners**
- **Charging Station Operators**
- **System Administrator**

Secondary Stakeholders

- **Payment Gateway Provider**
- **Government Authorities**
- **Maintenance Engineers**
- **EV Charging Network Companies**

Existing Challenges

- **Difficulty locating nearby charging stations.**
- **Lack of real-time charger availability.**
- **Long waiting times.**
- **Manual booking process.**
- **Payment issues.**
- **Poor monitoring of charging sessions.**
- **Inefficient station management.**

System Objectives

- **Provide user registration and secure login.**

- **Display nearby charging stations.**
 - **Enable online slot booking.**
 - **Provide secure online payment.**
 - **Monitor charging status in real time.**
 - **Manage charging stations efficiently.**
 - **Generate reports for administrators.**
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2. Functional and Non-Functional Requirements

Functional Requirements

User Module

- **User Registration**
- **User Login**
- **Forgot Password**
- **User Profile Management**

Charging Station Module

- **Search charging stations**
- **View station details**
- **Check charger availability**
- **View charging prices**

Slot Booking Module

- **Book charging slot**
- **Cancel booking**
- **Modify booking**
- **View booking history**

Payment Module

- **Online payment**
- **Payment confirmation**
- **Invoice generation**
- **Payment history**

Charging Monitoring Module

- **Start charging**
- **Stop charging**
- **Monitor charging progress**

- **View charging completion**

Notification Module

- **Booking confirmation**
- **Charging completion notification**
- **Payment confirmation**
- **Reminder notification**

Admin Module

- **Manage users**
- **Manage charging stations**
- **Manage chargers**
- **Manage bookings**
- **Manage payments**
- **Generate reports**

Non-Functional Requirements

Performance

- **Page should load within 3 seconds.**
- **Search results displayed within 2 seconds.**

Security

- **Secure login authentication.**
- **Password encryption.**
- **HTTPS communication.**
- **Role-based access control.**

Reliability

- **99.9% system uptime.**
- **Automatic backup.**

Usability

- **Simple and user-friendly interface.**
- **Responsive design.**

Scalability

- **Support thousands of concurrent users.**
- **Easily add new charging stations.**

Maintainability

- **Modular architecture.**
- **Easy software updates.**

Availability

- **System available 24×7.**
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3. Actors and Use Cases

Primary Actors

- **EV User**
- **Administrator**

Secondary Actors

- **Payment Gateway**
 - **Charging Station Operator**
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Use Cases

EV User

- **Register**
 - **Login**
 - **Search Charging Station**
 - **View Station Details**
 - **Book Slot**
 - **Make Payment**
 - **Monitor Charging**
 - **View Booking History**
 - **Logout**
-

Administrator

- **Login**
- **Add Charging Station**
- **Update Charging Station**
- **Delete Charging Station**
- **Manage Users**
- **Manage Bookings**
- **Manage Payments**

- **Generate Reports**
- **Logout**

Charging Station Operator

- **Update Charger Status**
- **Monitor Chargers**
- **View Bookings**

Payment Gateway

- **Process Payment**
- **Verify Payment**
- **Send Payment Status**

Use Case Diagram (Text Representation)

```
+-----+  
| EV Charging System |  
+-----+
```

EV User ----->

```
| Register  
| Login  
| Search Station  
| Book Slot  
| Payment  
| Monitor Charging  
| Logout
```

Administrator ----->

```
| Login  
| Manage Users  
| Manage Stations  
| Manage Bookings  
| Reports
```


Station Operator ----->

| Update Charger

| View Booking

Payment Gateway ----->

| Payment Verification

4. Software Requirement Specification (SRS)

4.1 Introduction

The EV Charging Station Management System is designed to simplify charging station discovery, slot reservation, online payment, and charging monitoring. It provides a centralized platform for EV users and charging station administrators.

4.2 Problem Description

Current charging station management systems suffer from inefficient booking, poor station visibility, and lack of real-time monitoring. Users waste time waiting for available chargers and often encounter payment difficulties.

4.3 Scope of the System

The system enables:

- User Registration**
 - Login**
 - Search Charging Station**
 - Slot Booking**
 - Online Payment**
 - Charging Monitoring**
 - Booking History**
 - Admin Management**
 - Report Generation**
-

4.4 Functional Requirements

FR1

The system shall allow user registration.

FR2

The system shall authenticate users.

FR3

The system shall display nearby charging stations.

FR4

The system shall allow slot booking.

FR5

The system shall process online payments.

FR6

The system shall monitor charging sessions.

FR7

The system shall notify users.

FR8

The administrator shall manage charging stations.

FR9

The administrator shall generate reports.

4.5 Non-Functional Requirements

- **Fast response time**
 - **Secure transactions**
 - **High availability**
 - **User-friendly interface**
 - **Reliable system**
 - **Scalable architecture**
 - **Easy maintenance**
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4.6 Use Case Descriptions

Use Case 1

Use Case Name: Register User

Actor: EV User

Precondition

User is not registered.

Main Flow

- 1. Open registration page.**
- 2. Enter details.**
- 3. Submit form.**
- 4. Account created.**

Postcondition

User account is successfully created.

Use Case 2

Use Case Name: Book Charging Slot

Actor

EV User

Precondition

User logged in.

Main Flow

- 1. Search station.**
- 2. Select station.**
- 3. Choose slot.**
- 4. Confirm booking.**

Postcondition

Booking confirmed.

Use Case 3

Use Case Name: Online Payment

Actor

EV User

Main Flow

- 1. Choose payment.**
- 2. Enter payment details.**
- 3. Verify payment.**
- 4. Receive confirmation.**

Use Case 4

Use Case Name: Manage Charging Station

Actor

Administrator

Main Flow

- 1. Login**
- 2. Add station**
- 3. Update station**
- 4. Delete station**

4.7 Data Requirements

User Table

- User ID**
- Name**
- Email**
- Phone**
- Password**

Charging Station Table

- Station ID**
- Station Name**
- Location**
- Charger Type**
- Number of Chargers**
- Availability**

Booking Table

- Booking ID**
- User ID**
- Station ID**
- Date**
- Time**

- **Status**
-

Payment Table

- **Payment ID**
 - **Booking ID**
 - **Amount**
 - **Payment Method**
 - **Payment Status**
-

Charging Session Table

- **Session ID**
 - **Booking ID**
 - **Start Time**
 - **End Time**
 - **Energy Consumed**
 - **Status**
-

4.8 External Interface Requirements

User Interface

- **Login Page**
 - **Registration Page**
 - **Dashboard**
 - **Charging Station Search**
 - **Booking Page**
 - **Payment Page**
 - **Charging Status Page**
 - **Admin Dashboard**
-

Hardware Interface

- **Computer**
- **Smartphone**
- **Charging Machine**
- **GPS**

Software Interface

- **Web Browser**
- **Payment Gateway API**
- **Database**
- **Maps API**

Communication Interface

- **Internet**
- **HTTPS**
- **Email Notification**
- **SMS Notification**

4.9 System Features

- **Secure Login**
- **User Registration**
- **Search Charging Stations**
- **Live Availability**
- **Slot Booking**
- **Online Payment**
- **Charging Status Monitoring**
- **Booking History**
- **Notifications**
- **Admin Dashboard**
- **Report Generation**

5. Assumptions and Constraints

Assumptions

- **Internet connection is available.**
- **Users have smartphones or computers.**
- **Charging stations provide real-time status.**
- **Payment gateway is always operational.**
- **GPS location is available.**

- **Database server remains online.**

Constraints

Technical Constraints

- **Internet dependency.**
- **GPS accuracy.**
- **API limitations.**

Operational Constraints

- **Staff training required.**
- **Regular maintenance.**

Economic Constraints

- **Development cost.**
- **Cloud hosting charges.**
- **Payment gateway transaction fees.**

Legal Constraints

- **User data privacy regulations.**
- **Secure payment compliance.**
- **Government EV regulations.**

Time Constraints

- **Project completion within the academic semester.**
- **Testing and deployment within the scheduled timeline.**

6. Requirement Validation

Requirement Completeness

- **✓ User Registration**
- **✓ Login**
- **✓ Charging Station Search**
- **✓ Slot Booking**
- **✓ Payment**
- **✓ Charging Monitoring**
- **✓ Notifications**
- **✓ Reports**

- **✓ Admin Module**

Requirement Consistency

- **No conflicting requirements identified.**
- **Functional and non-functional requirements are aligned with the system objectives.**

Ambiguity Check

Potential ambiguities:

- **"Nearby" charging station should be defined using GPS location and distance (e.g., within a configurable radius).**
- **"Real-time" charger availability should specify the refresh interval.**

Redundancy Check

- **Duplicate booking functionality removed.**
- **Payment confirmation is handled within the payment module to avoid overlap.**

Missing Requirement Check

Additional recommended requirements:

- **Password reset functionality.**
- **User profile management.**
- **Feedback and rating for charging stations.**
- **Emergency contact/support.**
- **Admin activity logs.**
- **Multi-language support.**
- **EV charging analytics dashboard.**

Suggested Improvements

- **Integrate AI-based charger recommendations based on user preferences.**
- **Add QR code-based charging session initiation.**
- **Enable push notifications through a mobile app.**

- **Support multiple payment methods (UPI, cards, wallets, net banking).**
- **Add predictive maintenance alerts for charging stations.**
- **Implement role-based access control for administrators and operators.**
- **Provide advanced analytics and reports on charger utilization and revenue.**

Conclusion:

The proposed EV Charging Station Management System provides a comprehensive solution for EV users and charging station administrators by enabling efficient station discovery, slot booking, secure payments, real-time charging monitoring, and centralized management. The identified requirements, use cases, assumptions, constraints, and validation ensure that the system is complete, consistent, scalable, and suitable for real-world deployment while satisfying the requirements of a professional Software Requirement Specification (SRS).